

IEEE

Through the Logic Gates

MULTI DOOR SAFETY INTERLOCK SYSTEM

Capturing the Art of
Electronic Components

Presented By: **Future Flow**



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Turning Gadgets Into Visual Stories



CAR SAFETY



PROBLEM STATEMENT

Accidents can occur if vehicles door are not properly closed.

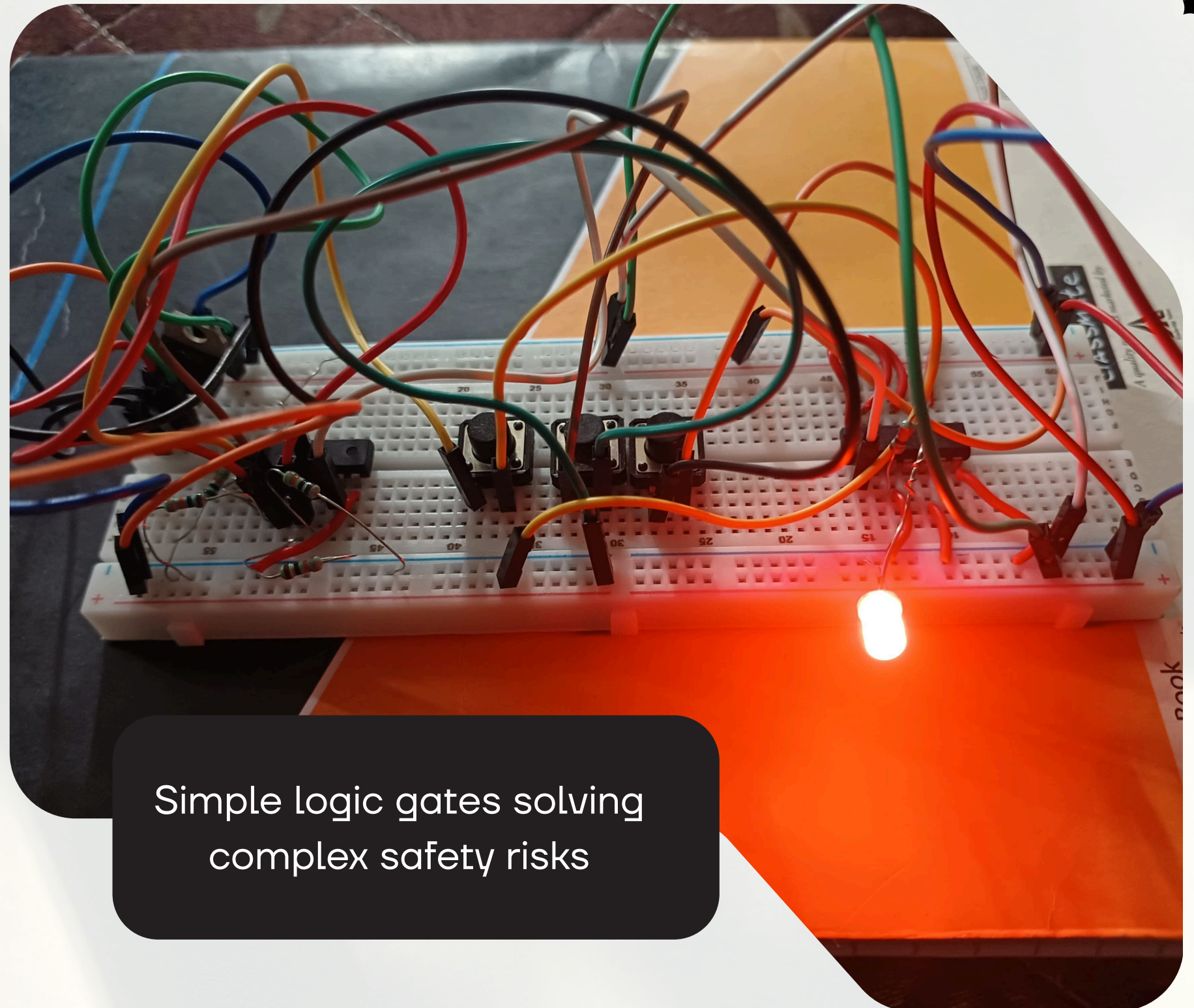
We need a reliable system to prevent engine start in unsafe condition.



ABOUT THE PROJECT

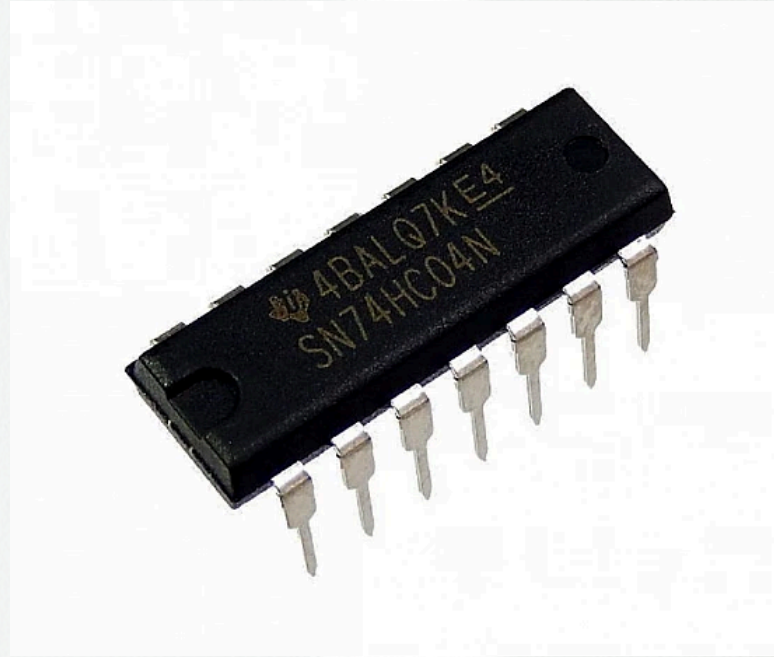
This Project ensures that a vehicle engine starts only when all doors are closed

It uses digital logic (AND and NOT gates) for safety interlocking

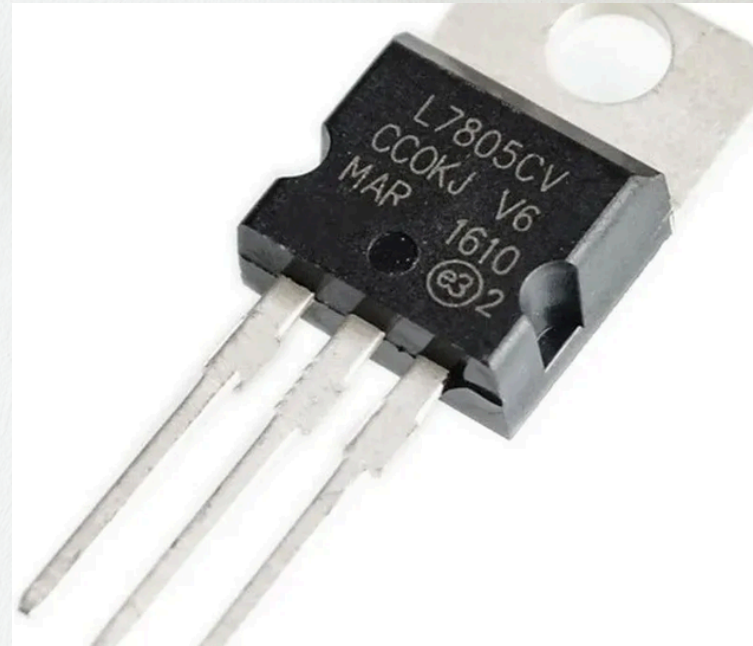


Simple logic gates solving complex safety risks

COMPONENTS



74HC04 IC



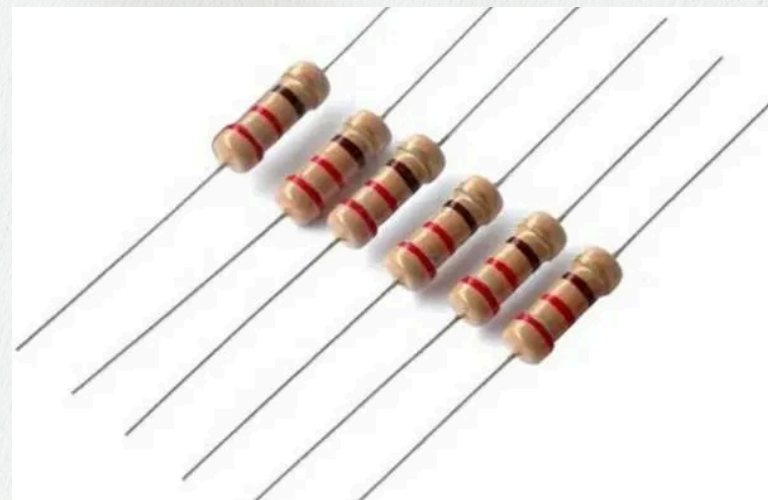
7805 Regulator



74HC08 IC



Push Button



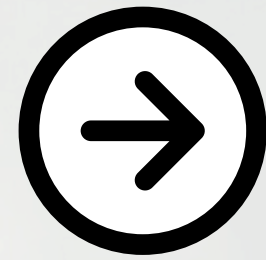
Resistors



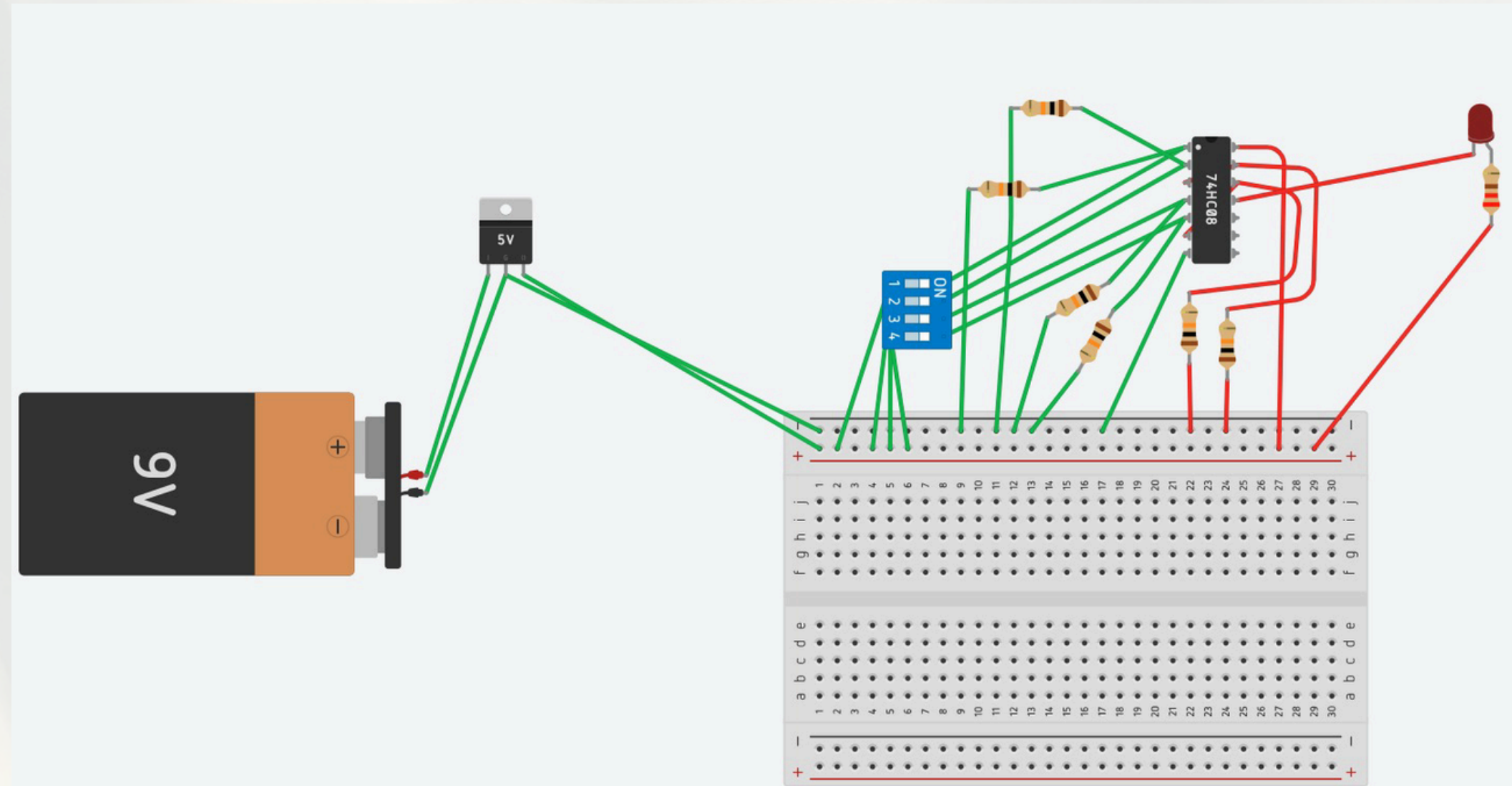
Led Light

BEHIND THE PROJECT

CIRCUIT DIAGRAM



Ensuring automotive safety through deterministic digital logic constraints and hardware-level verification.



HOW IT WORKS



POWER ON

9V Battery
Powers the
circuit
through 7805
Regulator



INPUT (DOORS)

Push Buttons
act as door
inputs
(1 = closed
0 = opens)



NOT GATE

NOT gate
inverts the
input.
(open = 0
closed = 1)



AND GATE

AND gate
combines
the signal



OUTPUT CHECK

LED
glowes
only if all
the doors
are closed

RESULTS

The system works as expected

System enabled only when all doors are closed

If any door is open , Output is disabled

Instant response and Highly Reliable

FUTURE SCOPE

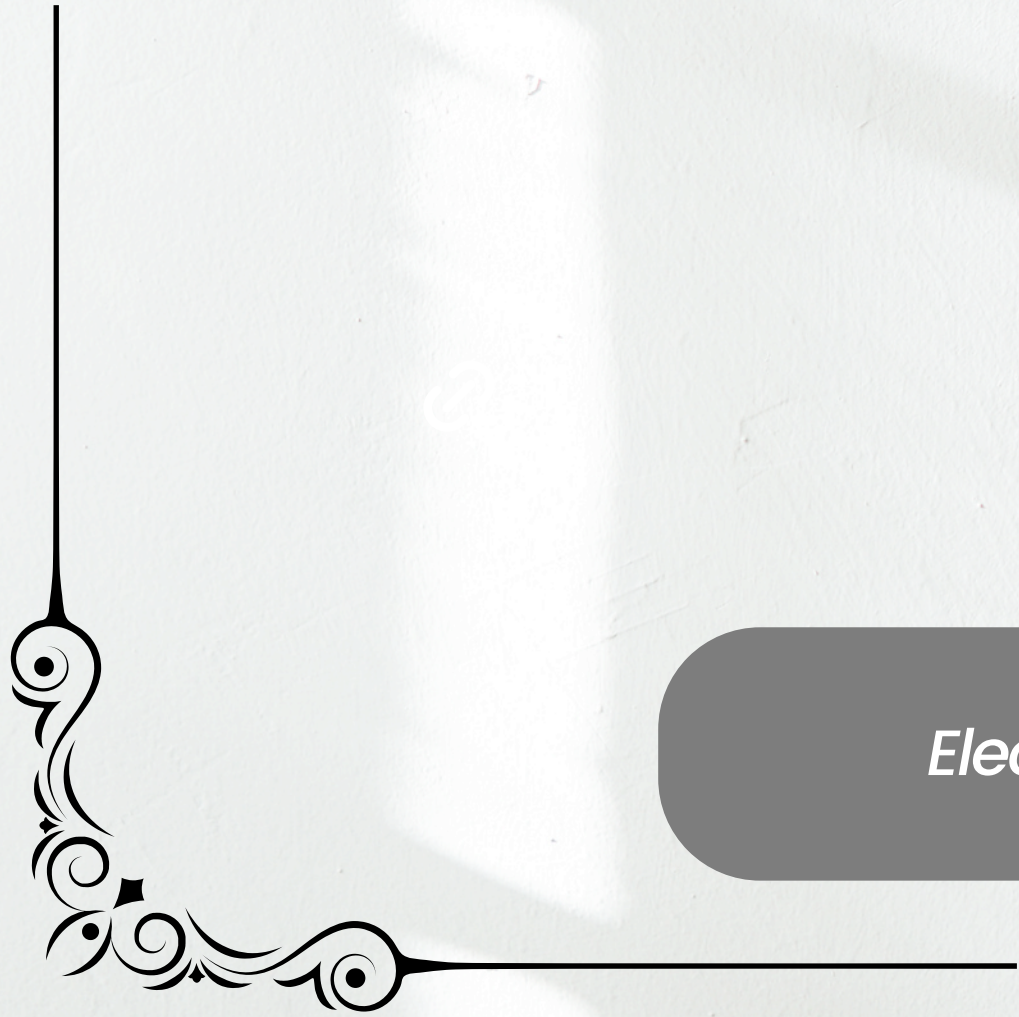
Can be extended with door sensors instead of push buttons

Integration with Buzzer Alarm for warning

Can be implemented in real vehicles for enhanced safety



THANK YOU SO MUCH



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